CS486 MOCK FINAL EXAMINATION - TEST 5

Based on the FINAL REVIEW structure; total: 10 points

Question 1 - ER Models [1 pt]

A research conference manages **Conferences**, **Tracks**, **Papers**, **Authors**, and **Reviews**. Conference acronyms are unique. Track names are unique only within a conference. Paper IDs are globally unique and each paper is submitted to exactly one track. Papers have one or more authors with an author order. A reviewer may review many papers; each paper receives reviews from multiple reviewers, and each review stores a score and confidence.

Draw an ER diagram with keys/cardinalities and indicate the key of Track and the associative Review relation after mapping.

Question 2 - Query Languages [2 pt]

Schemas:

$Paper(PID, Title, Conf, Track)$, $Author(AID, Name)$, $Writes(AID, PID, AuthorOrder)$, $Review(RID, PID,$

Write:

(a) RA to find papers with at least two distinct authors.

(b) TRC to find authors all of whose submitted papers have average review score at least 3. (You may state that aggregation is handled by an extended-calculus aggregate.)

(c) SQL to find, for each conference, the paper(s) with the highest average review score among papers having at least 3 reviews. Keep ties.

Question 3 - Database Constraints [1 pt]

Business rule: *A reviewer may not review a paper if that reviewer is an author of the same paper; review Score must be between 1 and 5 and Confidence between 1 and 3.*

Give row-level SQL CHECK constraints and a formal cross-table constraint plus trigger/assertion logic for the conflict-of-interest rule.

Question 4 - Serialization & Concurrency [2 pt]

Given

$$S = r_1(A)\ w_1(B)\ r_2(B)\ w_2(C)\ r_3(C)\ w_3(A)\ r_2(A)\ w_1(C).$$

(a) Build the precedence graph and decide conflict serializability.

(b) With commit order c_2, c_1, c_3, determine recoverability/cascadelessness if each read sees the latest preceding write on that item.

(c) Under strict 2PL, which locks must be held until commit/abort, and what anomaly does strictness prevent?

Question 5 - Recovery [1 pt]

A REDO-only system has:

```
<start ckpt (T1)>
<T1, A, 100>
<commit T1>
<end ckpt>
<T2, B, 200>
<T3, C, 300>
<commit T2>
<T3, D, 400>
--- crash ---
```

State which updates are redone and which are ignored. Explain the role of the completed checkpoint in limiting log scanning.

Question 6 - Functional Dependencies & Normalization [1 pt]

Let $R(A, B, C, D, E, F)$ with

$$F = \{AB \to C,\ C \to A,\ B \to D,\ D \to E,\ E \to F\}.$$

(a) Find all candidate keys.

(b) Determine whether $B \to D$ violates 3NF and/or BCNF.

(c) Give a lossless decomposition that removes the transitive chain $B \to D \to E \to F$ from the relation containing paper-like core attributes A, B, C.

Question 7 - Concurrency Control Execution [1 pt]

A table row stores $Seats = 1$.

```
T1: r(Seats); if Seats>0 then Seats:=Seats-1; w(Seats)
T2: r(Seats); if Seats>0 then Seats:=Seats-1; w(Seats)
```

Construct a lost-update interleaving. State the final stored Seats value, the real-world inconsistency, and one isolation mechanism that prevents it.

Question 8 - Query Optimization [1 pt]

Review has 3,000,000 rows, 100 rows/page (30,000 pages). ReviewerID has 30,000 distinct values. Score has 5 values. There is an unclustered B+ tree on ReviewerID (height 3), and a hash index on ReviewerID. Estimate expected matches for one ReviewerID and compare B+ tree, hash, and full scan for equality. Then explain why neither unclustered index is necessarily attractive for $Score = 5$ if 20% of the table qualifies.

– *End of Mock Test* –